LOG AGGREGATION AND ANALYSIS

MPCS 56550 INTRODUCTION TO DEVOPS (WINTER 2026)

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WHAT IS LOG AGGREGATION?

collecting logs from many services/hosts

converting logs into a consistent structured format

adding contextual metadata to logs (e.g., service/route name)

storing so you can query

it turns 'grep on one machine' into 'search across everything'.

WHAT IS LOG ANALYSIS?

SEARCHING AND FILTERING LOGS TO INVESTIGATE ISSUES

IDENTIFYING PATTERNS AND TRENDS

VISUALIZING METRICS AND DASHBOARDS

WHY THIS MATTERS IN DEVOPS?

- faster debugging (lower MTTR)
- traceability (track a single user request across multiple logs)
- incident forensics (what happened, when, where)
- performance insights (latency, error rates)
- compliance/audit trails (prove what happened in a system)
- alerting (e.g., unusual activity / service failures)

FROM "TRADITIONAL LOGS" TO MODERN LOG PIPELINES

Early systems (local logs):

- applications write logs to local files (no structure)
- engineers SSH into servers and inspect local log files
- logs scattered across many machines

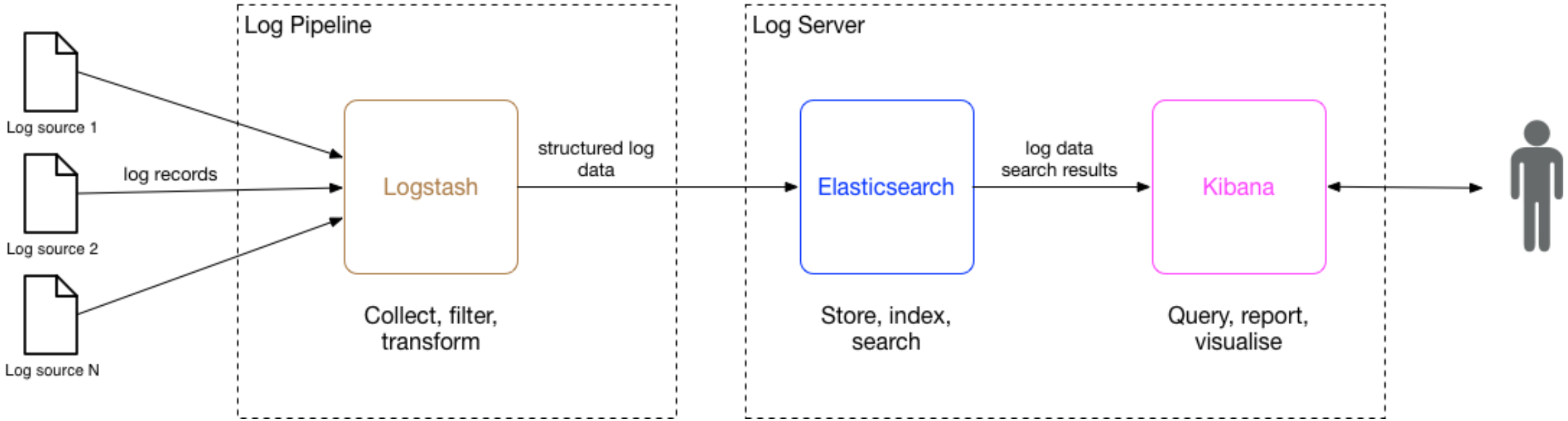
Modern:

- structured logs (json)
- logs (from diff. services) shipped to centralized systems
- indexed + searchable
- dashboards

CORE COMPONENTS

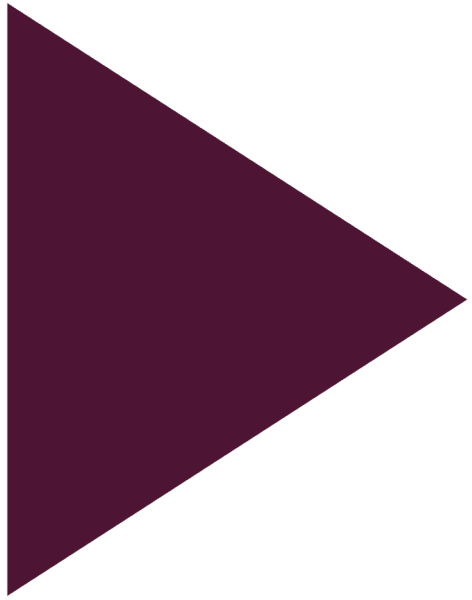
- producers: apps, services
- transport: tcp/udp, http, agents
- processing: parse, normalize, add metadata (logstash)
- storage/index: searchable store (elasticsearch)
- consumers: kibana

ARCHITECTURE (ELK STACK)



ELK STACK (POPULAR OPEN-SOURCE PLATFORM)

- Elasticsearch – distributed search and analytics engine
- Logstash – log ingestion and processing pipeline
- Kibana – visualization and dashboard interface



DEMO

github.com/abhyas01/log-aggregation-and-analysis/

REAL-WORLD IMPLEMENTATION

1. Large-scale web platforms (e.g., Netflix / Uber)
   - aggregate logs from thousands of microservices
   - quickly detect system failures
   - analyze traffic and error patterns

2. E-commerce platforms (e.g., Amazon / Shopify)
   - monitor checkout failures and payment errors
   - analyze user behavior and request patterns
   - investigate incidents

3. Security and compliance monitoring
   - track login activity and suspicious access
   - maintain audit trails of system actions
   - support security investigations

BENEFITS & CHALLENGES

Benefits

- centralized visibility across systems
- faster debugging and incident response
- powerful search across large volumes of logs
- real-time monitoring and alerting
- supports security auditing and compliance

Challenges

- large storage requirements for logs
- high indexing cost at scale
- noisy or excessive logging
- ensuring sensitive data is not logged

BEST PRACTICES

- use structured logging (json)
- include consistent fields (service, environment, request_id)
- log at appropriate levels (info / warn / error)
- avoid logging sensitive data (passwords, tokens)

COMMON PITFALLS

01

logging too much unnecessary data

02

missing timestamps or correlation IDs

03

unbounded log storage growth

EMERGING DEVELOPMENTS

AI/ML BASED LOG ANALYSIS TO DETECT ANOMALIES

COST-OPTIMIZED LOG STORAGE TIERS

INTEGRATION WITH DEVOPS PRACTICES

- ci/cd pipelines (monitor deployments)
- container orchestration platforms (kubernetes logging)
- security monitoring systems

QUESTIONS?